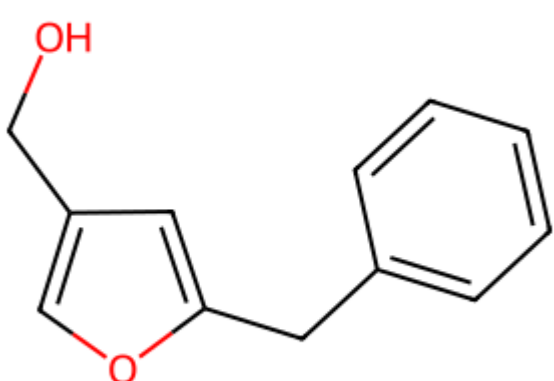


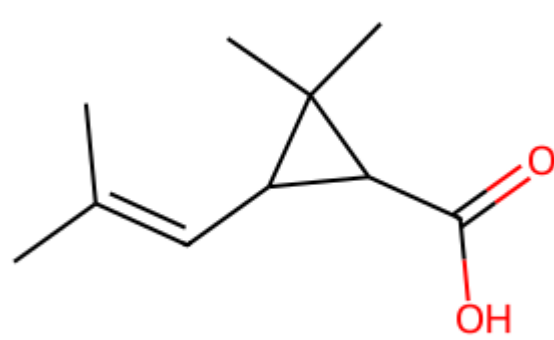
{'analysis\_time': 0.0,  
'first\_solution\_time': 0.593923,  
'num\_calls\_rxn\_model': 0,  
'num\_calls\_value\_function': 0,  
'reaction\_number\_estimate': 0.0,  
'retro\_star\_min\_cost': 4.896229956444593,  
'retro\_star\_mol\_cost': inf,  
'retro\_star\_reaction\_number': 4.896229956444593,  
'retro\_star\_rxn\_cost': 0.4812668215244463,  
'retro\_star\_value': 4.896229956444593}

Reaction

{'analysis\_time': 0.224048,  
'first\_solution\_time': 0.593923,  
'num\_calls\_rxn\_model': 1,  
'num\_calls\_value\_function': 0,  
'retro\_star\_min\_cost': 4.896229956444593,  
'retro\_star\_reaction\_number': 4.896229956444593,  
'retro\_star\_rxn\_cost': 0.4812668215244463,  
'retro\_star\_value': 4.896229956444593}



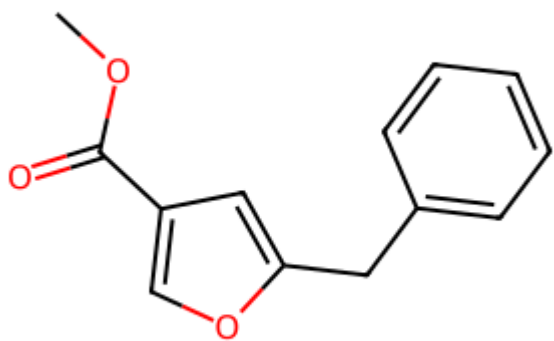
{'analysis\_time': 0.224048,  
'first\_solution\_time': 0.593923,  
'num\_calls\_rxn\_model': 1,  
'num\_calls\_value\_function': 0,  
'reaction\_number\_estimate': 0.0,  
'retro\_star\_min\_cost': 4.414963134920146,  
'retro\_star\_mol\_cost': inf,  
'retro\_star\_reaction\_number': 4.414963134920146,  
'retro\_star\_value': 4.896229956444593}



{'analysis\_time': 0.224048,  
'first\_solution\_time': 0.224048,  
'num\_calls\_rxn\_model': 1,  
'num\_calls\_value\_function': 0,  
'retro\_star\_min\_cost': 0.0,  
'retro\_star\_mol\_cost': 0.0,  
'retro\_star\_reaction\_number': 0.0,  
'retro\_star\_value': 4.896229956444593}

Reaction

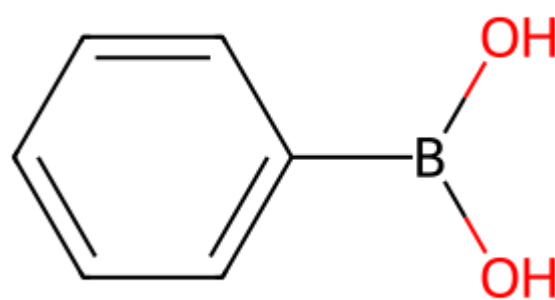
{'analysis\_time': 0.341687,  
'first\_solution\_time': 0.593923,  
'num\_calls\_rxn\_model': 2,  
'num\_calls\_value\_function': 54,  
'retro\_star\_min\_cost': 4.414963134920146,  
'retro\_star\_reaction\_number': 4.414963134920146,  
'retro\_star\_rxn\_cost': 1.2207799226423173,  
'retro\_star\_value': 4.896229956444593}



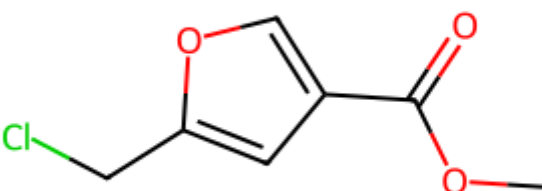
{'analysis\_time': 0.341687,  
'first\_solution\_time': 0.593923,  
'num\_calls\_rxn\_model': 2,  
'num\_calls\_value\_function': 54,  
'reaction\_number\_estimate': 0.0,  
'retro\_star\_min\_cost': 3.194183212277829,  
'retro\_star\_mol\_cost': inf,  
'retro\_star\_reaction\_number': 3.194183212277829,  
'retro\_star\_value': 4.896229956444593}

Reaction

{'analysis\_time': 0.593923,  
'first\_solution\_time': 0.593923,  
'num\_calls\_rxn\_model': 4,  
'num\_calls\_value\_function': 151,  
'retro\_star\_min\_cost': 3.194183212277829,  
'retro\_star\_reaction\_number': 3.194183212277829,  
'retro\_star\_rxn\_cost': 3.194183212277829,  
'retro\_star\_value': 4.896229956444593}



{'analysis\_time': 0.593923,  
'first\_solution\_time': 0.593923,  
'num\_calls\_rxn\_model': 4,  
'num\_calls\_value\_function': 151,  
'retro\_star\_min\_cost': 0.0,  
'retro\_star\_mol\_cost': 0.0,  
'retro\_star\_reaction\_number': 0.0,  
'retro\_star\_value': 4.896229956444593}



{'analysis\_time': 0.593923,  
'first\_solution\_time': 0.593923,  
'num\_calls\_rxn\_model': 4,  
'num\_calls\_value\_function': 151,  
'retro\_star\_min\_cost': 0.0,  
'retro\_star\_mol\_cost': 0.0,  
'retro\_star\_reaction\_number': 0.0,  
'retro\_star\_value': 4.896229956444593}